SACHA M. O. **SIANI**

PHD STUDENT • MSC REMOTE SENSING • BSC ENVIRONMENTAL AND SANITARY ENGINEERING

Department of Geography, Indiana University, Student Building 331, 701 E. Kirkwood Avenue, Bloomington, IN, USA 47405
ssiani@indiana.edu | +1 (812) 606–9754 | sachasiani.github.io

Sacha Siani is a PhD student in the Department of Geography at Indiana University, Bloomington. Siani holds Master's degree in Remote Sensing from National Institute for Space Research (INPE), Brazil (2016) and bachelor's degree in Environmental and Sanitary Engineering from University of Taubaté (2012). His research focuses on the effects of conservation policies on deforestation and forest degradation, as well as their social and economic impacts, in the Brazilian Amazon.

EDUCATION

Aug. 2018 – Present — *Bloomington, IN, USA*
PhD in Geography, Indiana University

Mar. 2014 – Aug. 2016 — *São José dos Campos, SP, Brazil*
MSc in Remote Sensing, National Institute for Space Research **–** INPE

Jul. 2013 – Not completed — *Taubaté, SP, Brazil*
MBA in Project Management, Getulio Vargas Foundation – FGV

Mar. 2008 – Nov. 2012 — *Taubaté, SP, Brazil*
BSc in Environmental and Sanitary Engineering, University of Taubaté – UNITAU

SKILLS

Languages	Portuguese (native), English (advanced), Spanish (basic)
Programming	R, Python, GoogleEarthEngine (JavaScript), MatLab, LaTeX
Softwares	QGis, ArcGis, TerraView, E–Cognition, Envi, Erdas, Weka, Adobe Illustrator, Adobe InDesign

RESEARCH EXPERIENCE

Aug. 2016 – Aug. 2018 — *São José dos Campos, SP, Brazil*
Research Assistant, Norwegian School of Economics – NHH

Project: Tropical Deforestation and Economic Development
Advisors: Dr. Torfinn Harding (NHH) and Dr. Liana O. Anderson (CEMADEM / INPE)

- Aims to understand the effects of environmental policies and economic development on tropical deforestation in the Brazilian Amazon;
- Empirical approaches based on spatial data analysis and econometrics;
- Tidied a big geographic database and performed statistical analysis using R;
- Used the Google Earth Engine platform to process time series of remote sensing data;
- Studied the effects of environmental policies such as increased law enforcement and protected areas on outcomes such as deforestation and forest fires;
- Analyzed the actor–specific forest cover dynamics;
- Participated in a field trip to the Brazilian Amazon.

Mar. 2014 – Aug. 2016 *São José dos Campos, SP, Brazil*

Master Student, National Institute for Space Research – INPE

Project: Socioenvironmental indicators for monitoring the Mananciais do Rio Paraíba do Sul Protected Area with Sustainable Use
Advisors: Dr. Silvana Amaral and Dr. Antônio Miguel V. Monteiro

- Developed a spatially explicit indicator system to assist the socioeconomic and environmental diagnostics and monitoring of a specific protected area with sustainable use;
- Created a novel conceptual framework for developing indicators systems for monitoring and the territorial planning of protected areas;
- Performed a land cover classification based on Landsat images and a Geographic–Object–Based Image Analysis approach using E–Cognition software;
- Applied spatial and statistical analysis using QGIS and R.

Jan. 2012 – Dec. 2012 *Taubaté, SP, Brazil*

Undergraduate Student, Department of Engineering, University of Taubaté – UNITAU

Project: Technical studies to support the creation of the Vale do Itaim Municipal Park in Taubaté–SP
Advisor: Dr. Ademir Fernando Morelli

- Characterized the socioeconomic and cultural aspects of the municipality of Taubaté using census data and historical documents;
- Applied descriptive data analysis and graphical representation.

Sep. 2009 – Dec. 2009 *Ubatuba, SP, Brazil*

Research Assistant, Atmosfera Enterprises

Project: Socioenvironmental diagnostic of the Saco da Ribeira district in Ubatuba–SP
Advisor: Dr. Patrícia Ortiz Monteiro

- Surveyed the sewage disposal, water supply and garbage collection, to diagnose a neighborhood where a marina would later be built;
- Participated in a field survey that involved the application of questionnaires.

TEACHING EXPERIENCE

Aug 2018 – Present *Taubaté, SP, Brazil*

Associate Instructor, Department of Geography at Indiana University

- I am AI in the Mapping Our World course, which is basically an introduction to Geographic Information Science and Cartography;
- This is an undergraduate level course.

May 2016 and Oct. 2017 *Taubaté, SP, Brazil*

Guest Instructor of Geoprocessing, University of Taubaté – UNITAU

- The course of geoprocessing applied to beekeeping was taught in the undergraduate program on Beekeeping;
- It was a four hours course and included an introduction to geoprocessing and a laboratory using the software QGis.

Jul. 2017 *São José dos Campos, SP, Brazil*

Guest instructor of introduction to R applied to spatial data, National Institute for Space Research – INPE

- The public were graduate students in remote sensing at INPE, as a topic of Population, Space and Environment course;
- It was a four hours course and included an introduction to R and to spatial data analysis.

ADDITIONAL EXPERIENCE

Aug. 2016 – Nov. 2016 *São Paulo, SP, Brazil*

GIS and remote sensing analyst, Siaa Consulting

- Evaluated the extent of the biggest environmental disaster in Brazil, the disruption of the mining tailing dam in Mariana–MG;
- Mapped the loss of forest cover caused by mudslides.

May 2011 – Dec. 2012 *Pindamonhangaba, SP, Brazil*

Environmental Engineering Intern, Tenaris Confab Industrial

- Multinational company in the steel industry;
- Collaborated to improve and maintain the Environmental Management System based on the ISO 14001:2004 standard;
- Performed environmental education activities with employees;
- Acquired organizational and interpersonal skills.

Jan. 2012 – Nov. 2012 *Ubatuba, SP, Brazil*

Environmental and Sanitary Engineering Intern, Gondwana Institute

- The project involved actions of sanitation, education and health in the traditional fishing community of Almada in Ubatuba–SP;
- Trained the local community and supported the community planning for the tourism season.

Aug. 2011 – Sep. 2011 *Bananal, SP, Brazil*

Environmental Engineering Intern, Forest Service of the State of São Paulo

- Aimed to create a socioeconomic diagnosis for the management plan of Strict Nature Reserve of Bananal;
- Organized and analyzed census data on socioeconomics.

PUBLICATIONS

Journal Articles

Omachi, C.Y., **Siani, S.M.O.**, Chagas, F., Mascagni, M., Cordeiro, M., Garcia, G., Thompson, C.C., Siegle, E., Thompson, F.L., 2018. Atlantic Forest loss caused by the world´s largest tailing dam collapse (Fundão Dam, Mariana, Brazil). Remote Sens. Appl. Soc. Environ. doi:10.1016/j.rsase.2018.08.003

Siani, S.M.O., Amaral, S., Monteiro, A.M.V., 2017. iSAM – Um sistema de indicadores para o monitoramento da Área de Proteção Ambiental Mananciais do Rio Paraíba do Sul. Rev. do Dep. Geogr. 33, 63. doi:10.11606/rdg.v33i0.119663

Conference Proceedings

França, D.G.M., Anderson, L.O., Rocha, L., **Siani, S.M.O.**, Maia, M.R. da S.A., 2017. Random forest classification applied to sand pit detection in Cruzeiro do Sul, Acre, Brazilian Amazon, in: XVIII Brazilian Symposium on Remote Sensing. Galoa, Santos, pp. 8000–8007.

Moura, Y., Galvão, L., Anderson, L., **Siani, S.**, Aragão, L., 2017. Use of Green–Red Normalized Difference (GRND) index to evaluate large–scale canopy phenology in tropical forests, in: EGU General Assembly 2017. Vienna, p. 10162.

Pessôa, A.C.M., **Siani, S.M.O.**, Harding, T., Anderson, L.O., 2017. The role of Protected Areas on active fires occurrence: an exploratory study in Acre state, Brazilian Amazon, in: XVIII Brazilian Symposium on Remote Sensing. Galoa, Santos, pp. 7382–7382.

Pletsch, M.A.J.S., Körting, T.S., Escada, M.I.S., **Siani, S.M.O.**, 2017. Data mining applied to temporal dynamics of deforestation pattern: a study case in Southern Amazon forest, Brazil, in: XVIII Brazilian Symposium on Remote Sensing – SBSR. Santos, pp. 6959–6966.

* **Siani, S.M.O.**, Pessôa, A.C.M., Harding, T., Anderson, L.O., 2017. Spatial analysis of enforcement for reducing deforestation in the Brazilian Amazon: an exploratory study in Pará state, in: XVIII Brazilian Symposium on Remote Sensing – SBSR. Galoa, Santos, pp. 7451–7458.

França, D.G.M., **Siani, S.M.O.**, Moraes, E.C. De, 2015. Análise da correção atmosférica por FLAASH sobre cenas OLI Landsat–8, in: XVII Brazilian Symposium on Remote Sensing – SBSR. João Pessoa, pp. 6526–6533.

Siani, S.M.O., Campos, J., França, D.G.M., Amaral, S., Monteiro, A.M.V., Körting, T.S., 2015a. Land–cover classification of an intra–urban environment using high–resolution images and geographic object–based image analysis: the case of APA Mananciais do Rio Paraíba do Sul, in: Simpósio Brasileiro de Sensoriamento Remoto – SBSR. Instituto Nacional de Pesquisas Espaciais – INPE, João Pessoa, pp. 997–1004.

Siani, S.M.O., Escada, M.I.S., Amaral, S., Monteiro, A.M.V., 2015b. A dinâmica dos padrões de cobertura urbana na Área de Proteção Ambiental Mananciais do Rio Paraíba do Sul, in: Simpósio Brasileiro de Sensoriamento Remoto – SBSR. Instituto Nacional de Pesquisas Espaciais – INPE, João Pessoa, p. 8.

Siani, S.M.O., Amaral, S., Monteiro, A.M.V., 2014. A Sustentabilidade e seu Arranjo Espacial no Município de Ubatuba, Litoral Norte de São Paulo: a construção de um indicador de desenvolvimento sustentável, in: Encontro Nacional de Estudos Populacionais. ABEP, São Pedro, SP, Brasil, p. 15.

Siani, S.M.O., Monteiro, P.O., 2010. Diagnóstico Socioambiental do Bairro do Saco da Ribeira, Ubatuba–SP, in: XV Encontro de Iniciação Científica Da Universidade de Taubaté. Taubaté.

Master Thesis

Siani, S.M.O. iSAM – Um sistema de indicadores para o monitoramento da Área de Proteção Ambiental Mananciais do Rio Paraíba do Sul. Thesis (Master in Remote Sensing). National Institute for Space Research – INPE, 2016, p. 191.

Working Papers

Bento, B.P., **Siani, S.M.O.**, Pessoa, A.C.M., Aragão, L.E.O.C., Harding, T., Anderson, L.O., 2018. Deforestation and forest cover dynamics in rural properties in the Brazilian Amazon. São José dos Campos.

Pessôa, A.C.M., **Siani, S.M.O.**, Bento, B.P., Harding, T., Aragão, L.E.O.C., Anderson, L.O., 2018. Reducing environmental impacts through land regulation policy: evidence from the Brazilian Amazon fires.

Siani, S.M.O., Pessôa, A.C.M., Harding, T., Aragão, L.E.O.C., Anderson, L.O., 2018. The effects of policing on deforestation in Brazilian Legal Amazon. São José dos Campos.

Vedovato, L.B., **Siani, S.M.O.**, Wagner, F.H., Gesteira, M., Arai, E., Anderson, L.O., Aragão, L.E.O.C., 2018. Forest fragmentation relations with fire occurrence in the Brazilian Amazon.

Presentations

Bento, B.P., **Siani, S.M.O.**, Harding, T., Pessoa, A.C.M., Anderson, L.O., 2018. Deforestation and forest cover dynamics in rural properties in the Brazilian Amazon, in: Workshop on Policies Against Tropical Deforestation. Oslo, Norway.

Siani, S.M.O., Harding, T., Pessoa, A.C.M., Bento, B.P., Anderson, L.O., 2018. Satellite–based Monitoring and Law–Enforcement to Curb Deforestation in the Brazilian Amazon, in: Workshop on Policies Against Tropical Deforestation. Oslo, Norway.

Bento, B.P., **Siani, S.M.O.**, Pessoa, A.C.M., Harding, T., Anderson, L.O., 2017. Actor–specific contribution to deforestation in the Brazilian Amazon, in: 5th Bergen Economics of Energy and Environment Research Conference – BEEER. Bergen, Norway.

Pessôa, A.C.M., **Siani, S.M.O.**, Harding, T., Anderson, L.O., 2017. The impacts of land regulation policy on fire occurrence: an exploratory study in Acre state, Brazilian Amazon, in: 5th Bergen Economics of Energy and Environment Research Conference – BEEER. Bergen, Norway.

Siani, S.M.O., Pessoa, A.C.M., Harding, T., Anderson, L.O., 2017. Spatial Analysis of enforcement for reducing deforestation in the Brazilian Amazon: an exploratory study in Pará state, in: 5th Bergen Economics of Energy and Environment Research Conference – BEEER. Bergen, Norway.

Siani, S.M.O., Amaral, S., Monteiro, A.M. V, 2015a. Construção de um sistema de indicadores para o monitoramento da Área de Proteção Ambiental Mananciais do Rio Paraíba do Sul, in: V Seminário Da Linha de Pesquisa População, Ambiente, Espaço E Sustentabilidade E II Diálogos Transversais NEPO–INPE. São José dos Campos.

Siani, S.M.O., Amaral, S., Monteiro, A.M. V, 2015b. iSAM – Um sistema de indicadores para o monitoramento da Área de Proteção Ambiental Mananciais do Rio Paraíba do Sul, in: Diálogos Transversais – INPE, UFABC E USP – Geotecnologias Para Análise Do Território. São José dos Campos.

AWARD

2015	* **Best paper award**, XVII Brazilian Symposium on Remote Sensing – SBSR. João Pessoa, Brazil.
2018 – 2019	**Associate Instructor award for the 2018-2019 academic year**, Department of Geography at Indiana University. Bloomington, USA.
2019	**2019 Summer Research Fellowship**, Department of Geography at Indiana University. Bloomington, USA.

EXTRACURRICULAR ACTIVITY

PhD Summer School on Econometrics for Studies on Natural Resources and Development, Norwegian School of Economics – NHH	*18h* *May 2017*
PhD Summer School on Econometrics for Studies on Natural Resources and Development, Norwegian School of Economics – NHH	*45h* *Sep. 2016*
Training for the Rural Environmental Registry (CAR), Brazilian Forest Service	*16h* *Aug. 2015*
Introduction to Matlab, National Institute for Space Research – INPE	*20h* *May 2015*
Erdas Imagine Fundamentals I & II, Santiago & Cintra Consulting	*40h* *Sep. 2014*
ArcGIS I: Introduction to GIS & ArcGIS II: Essential Workflows, Imagem/ESRI	*40h* *May 2013*
Training of Internal Auditors ISO 14001:2004 & OHSAS 18001:2007, Tenaris University/SENAI	*24h* *Nov. 2014*
5S Methodology, Tenaris University	*4h* *Aug. 2012*
Environmental Management Tools, Tenaris University	*4h* *Nov. 2011*

Environment, Safety and Occupational Health, First Aid, Quality, Professional Behavior and Production Processes, Tenaris University/SENAI	*45h* *May 2011*
Environmental Licensing, Senac São Paulo	*40h* *May 2011*
Solid waste management, University of Taubaté – UNITAU	*4h* *May 2012*
Cultural Exchange in Germany, Rotary International	*2 months* *Jun. 2008*
Master English Course, CCAA English School	*2001 – 2006*